CCNA Exam Prep: Back to Networking Basics with Hank Preston and Patrick Gargano

A CCNA Diagnostic Playbook: Isolating Faults from Layer 1 to Layer 7

When network connectivity breaks, where do you begin? Troubleshooting is a cornerstone of the CCNA v2.0 exam, and having a systematic diagnostic methodology is what separates guessing from true engineering. In this live stream, Hank and Patrick build a practical diagnostic playbook from the ground up. We'll explore structured troubleshooting models using the OSI stack, highlighting core diagnostic tools, including `show` commands, syslog inspection, standard and extended ping, and traceroute. We'll also analyze pre-staged packet capture output to interpret real-world protocol behaviors (ARP, DHCP, and 802.1Q tags) without getting bogged down in capture setup. Join us to sharpen your troubleshooting instincts!

> Blueprint Topics:
>
> - 1.1: Diagnose interface and cable issues (collisions, errors, duplex, speed)
> - 1.6 Troubleshoot wired and wireless client connectivity (IP configuration, network reachability, and wireless security parameters on Windows, macOS, and Linux)
> - 2.4: Troubleshoot basic Layer 2/Layer 3 connectivity and device operations using show commands (including show logs), ping, extended ping, traceroute, and packet capture output
>
> Note: A PDF version of this lab guide is available on the CML Community.

Lab Requirements

This CML topology and lab require a CML server that supports public Internet access through the external connector named "Internet". It is configured in NAT mode to provide reachability to Internet resources through the CML host. This access is required to configure the "server" node and to complete some of the lab exercises.

If your CML server is "air-gapped" and doesn't have Internet access for topologies, this lab will not work as designed.

Setup and Scenario

This is a simplified network topology connecting a "client" and a "server" that is used as the basis for a series of "trouble tickets": scenarios where some network-based traffic is not working as expected.

The "server" is an Alpine Linux host, reachable at IP address 10.80.15.3. It acts as a DNS server for the domain "example.com" and provides recursive lookups to public DNS servers. It also represents a "web application server" hosting files that the "client" should be able to access.

The "client" is an Alpine Linux host and uses DHCP from "sw2" to configure its IP address. See `/etc/network/interfaces`.

Pre-validation

After starting the lab topology, verify the starting state by running these commands on "server".

> These commands verify that the startup configuration for "server" was completed successfully. This includes installing the DNS and HTTP services used in this lab.

```
server:~$ dig server.example.com

; <<>> DiG 9.20.29 <<>> server.example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23149
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 452d18fa4c4598ab010000006acfeb1709166434033ffc57 (good)
;; QUESTION SECTION:
;server.example.com.            IN      A

;; ANSWER SECTION:
server.example.com.     86400   IN      A       10.80.15.3

;; Query time: 1 msec
;; SERVER: 10.80.15.3#53(10.80.15.3) (UDP)
;; WHEN: Wed Oct 14 20:50:31 UTC 2026
;; MSG SIZE  rcvd: 91
```

```
server:~$ wget -O - http://server/test.txt
Connecting to server (127.0.0.1:80)
writing to stdout
Success!
-                    100% |********************************|     9  0:00:00 ETA
written to stdout
```

Trouble Tickets

The following troubleshooting tickets can be completed using the CML topology after it starts. Each ticket is self-contained; however, they should be completed in order because later tickets may "assume" that previous tickets have already been solved.

Troubleshooting walkthroughs for each ticket are provided, but initially collapsed/hidden so you can try to solve the ticket on your own.

Trouble Ticket #1: Client / Server Not Making a Connection

Title: A newly deployed "client" can't ping its "server"

Description: The team just deployed a new "client" host at a new site. The "client" is connected to "sw2" on port `e1/0`. It needs to be able to ping "server" (`10.80.15.3/29`) which is connected to "sw1". Both hosts are on the "APPLICATION" network.

Troubleshooting Method Suggestion: Bottom Up / Hop-by-Hop

Ticket Goal: Identify and fix the issue preventing "client" from pinging "server"

▶ **Troubleshooting Walkthrough**

Trouble Ticket #2: Secure Remote Management Not Working

Title: Unable to connect to "rtr1" with SSH for management

Description: An audit of the network's secure management protocols found that "rtr1" (`10.80.0.1`) is reachable via Telnet, but SSH is not working. SSH version 2 must be enabled and use local username and password authentication. Telnet must be disabled.

- Test access from "server" for this ticket.
- All credentials on "rtr1" are set to `cisco`

Troubleshooting Method Suggestion: Top-Down

Ticket Goal: Identify and resolve the problem with SSH management. Also disable Telnet access.

▶ **Troubleshooting Walkthrough**

Trouble Ticket #3: Client Configuration Download Failing

Title: Client unable to download config or data from "server"

Description: While setting up the application on "client", the systems team encountered an issue downloading the required files `http://server/config.file` and `http://server/data.file`. The test file `http://server/test.txt` does work.

- The command `wget -T 10` can be used to download files. Ex: `wget -T 10 http://server/test.txt`
  - Note: The `-T 10` argument sets the timeout for `wget` to 10 seconds. The default is indefinite
- Determine why the file downloads are failing on "client"

Troubleshooting Method Suggestion: Top-Down

Ticket Goal: Diagnose the problem. Determine the most likely cause and suggest ways to resolve it. Actually fixing the problem is not necessary.

▶ **Diagnostic Walkthrough**

▶ **Fixing the Issue!**

Trouble Ticket #4: Internet is down!

> Note: We got a little excited about the topic, and this lab guide got a little too big to embed within CML. You can find Ticket 4 in the PDF lab guide available with the resources on the CML Community.

Title: "Client" unable to access Internet websites

Description: The "client" can now access "server" resources; however, it can't reach any public Internet services, even though it should be able to.

> - Use "ping" to validate access to resources
> - You can validate Internet access using hosts `u.cisco.com` or `cisco.com`
> - If your lab environment blocks ICMP traffic to the Internet, you can use `csnake.example.com` and `quantumcloud.example.com` as alternatives. (These addresses point to the "server" host)

Troubleshooting Method Suggestion: Divide and Conquer + Packet Capture

Ticket Goal: Find and fix the problem preventing Internet access for "client".

▶ **Troubleshooting Walkthrough**